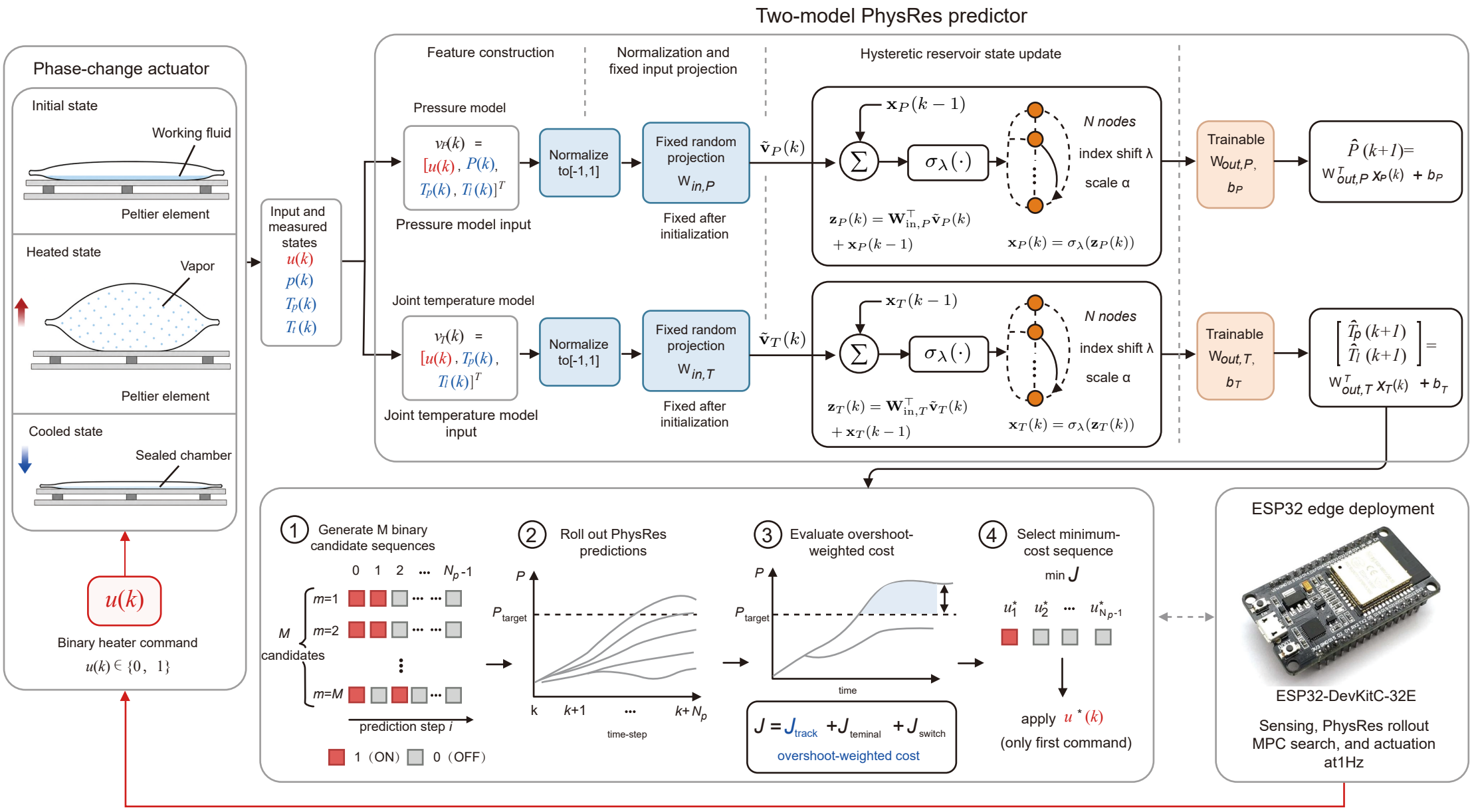

Two-model PhysRes predictor
Phase-change actuator
Initial state
Working fluid
Peltier element
Heated state
Vapor
Peltier element
Cooled state
Sealed chamber
u(k)
Binary heater command
u(k) ∈ {0, 1}
Input and measured states
u(k)
p(k)
Tp(k)
Ti(k)
Feature construction
Normalization and fixed input projection
Hysteretic reservoir state update
Pressure model
vP(k) = [u(k), P(k), Tp(k), Ti(k)]T
Pressure model input
Normalize to[-1,1]
Fixed random projection
Win,P
Fixed after initialization
ṽP(k)
xP(k − 1)
σλ(·)
N nodes
index shift λ
scale α
zP(k) = WTin,P ṽP(k) + xP(k − 1)
xP(k) = σλ(zP(k))
Trainable Wout,P, bP
P̂(k+1)= WTout,P xP(k) + bP
Joint temperature model
vT(k) = [u(k), Tp(k), Ti(k)]T
Joint temperature model input
Normalize to[-1,1]
Fixed random projection
Win,T
Fixed after initialization
ṽT(k)
xT(k − 1)
σλ(·)
N nodes
index shift λ
scale α
zT(k) = WTin,T ṽT(k) + xT(k − 1)
xT(k) = σλ(zT(k))
Trainable Wout,T, bT
[T̂p(k+1), T̂i(k+1)]T = WTout,T xT(k) + bT
1 Generate M binary candidate sequences
0 1 2 ... Np-1
m=1
m=2
M candidates
m=M
prediction step i
1 (ON) 0 (OFF)
2 Roll out PhysRes predictions
P
Ptarget
k k+1 ... k+Np
time-step
3 Evaluate overshoot-weighted cost
P
Ptarget
time
J = Jtrack + Jteminal + Jswitch
overshoot-weighted cost
4 Select minimum-cost sequence
min J
u*1 u*2 ... u*Np-1
apply u*(k)
(only first command)
ESP32 edge deployment
ESP32-DevKitC-32E
Sensing, PhysRes rollout MPC search, and actuation at1Hz